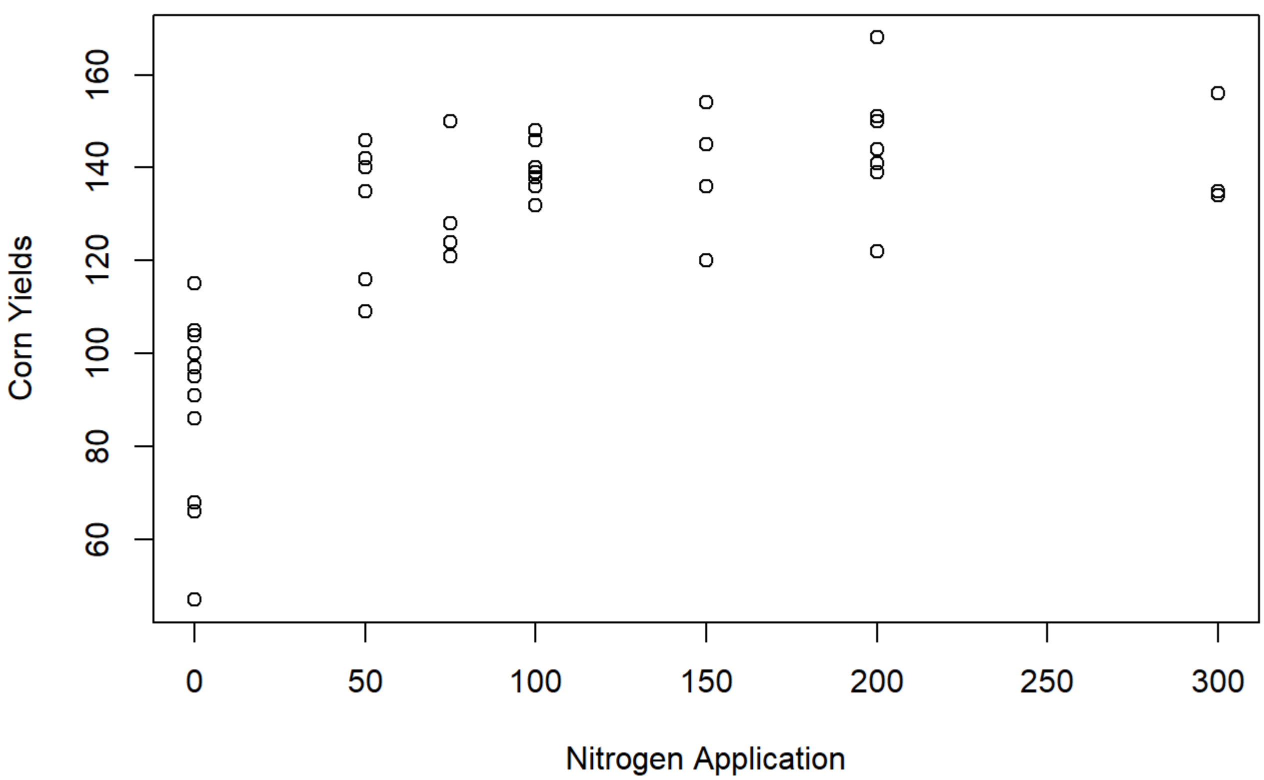
1

```r
corn_data = read.table("data/cornnit.txt", header=T)

head(corn_data)
```

```
  yield nitrogen
1   115        0
2   128       75
3   136      150
4   135      300
5    97        0
6   150       75
```

(i) (1 pt) What does the scatter plot of yields versus nitrogen suggest?

```r
plot(y=corn_data$yield, x=corn_data$nitrogen, xlab="Nitrogen Application", ylab="Corn Yields")
```



The scatter plot suggests a non linear relationship between nitrogen application and corn yields because the scatter plot does not form a straight line and is instead high slope at the start then starts to level off towards the end.

(ii) (1 pt) Fit separate linear first-order and second-order models with yield as the response and nitrogen as the predictor.

```
fit1 = lm(yield ~ nitrogen, data=corn_data)
summary(fit1)
```

```
Call:
lm(formula = yield ~ nitrogen, data = corn_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-60.439 -10.939   1.534  14.082  29.697 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 107.43864    4.66622   23.02  < 2e-16 ***
nitrogen      0.17730    0.03377    5.25 4.71e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 20.53 on 42 degrees of freedom
Multiple R-squared:  0.3962,	Adjusted R-squared:  0.3818 
F-statistic: 27.56 on 1 and 42 DF,  p-value: 4.713e-06
```

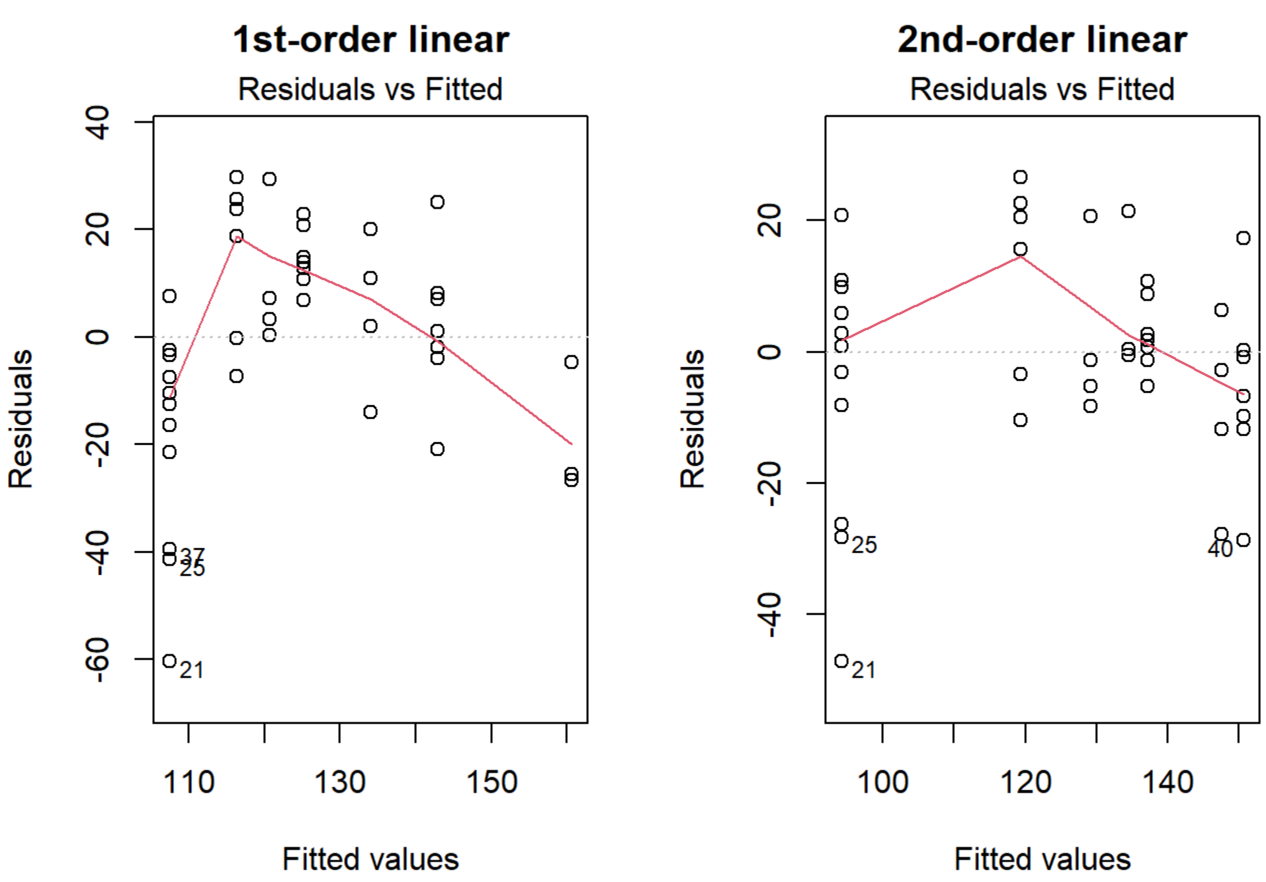
```
fit2 = lm(yield ~ nitrogen + I(nitrogen^2), data=corn_data)
summary(fit2)
```

```
Call:
lm(formula = yield ~ nitrogen + I(nitrogen^2), data = corn_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-47.162  -7.092   0.368  10.058  26.571 

Coefficients:
                Estimate Std. Error t value Pr(>|t|)    
(Intercept)   94.1620969  4.4067757  21.368  < 2e-16 ***
nitrogen       0.5794901  0.0800285   7.241 7.54e-09 ***
I(nitrogen^2) -0.0014831  0.0002787  -5.321 3.97e-06 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.98 on 41 degrees of freedom
Multiple R-squared:  0.6428,	Adjusted R-squared:  0.6254
F-statistic:  36.9 on 2 and 41 DF,  p-value: 6.817e-10
```

(iii) (2 pt) Is the second-order model a better fit?

```
par(mfrow=c(1,2))
plot(fit1, 1, main="1st-order linear")
plot(fit2, 1, main="2nd-order linear")
```




```
anova(fit1, fit2, test="F")
```

```
Analysis of Variance Table

Model 1: yield ~ nitrogen
Model 2: yield ~ nitrogen + I(nitrogen^2)
  Res.Df   RSS Df Sum of Sq      F    Pr(>F)    
1     42 17699                                  
2     41 10470  1    7229.6 28.312 3.974e-06 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

According to the anova test there is a significant relationship between yield and the extra predictor nitrogen^2. Thus, the second-order model is a better fit than the first-order model.

(iv) (2 pt) What can you conclude about the relationship between yield and nitrogen?

This the relationship between yield and nitrogen is non linear because the second-order model is a better fit since the nitrogen^2 predictor had a statistically significant relationship with the response variable yield.

2

```r
diabetes = read.csv("data/diabetes.csv")

head(diabetes)
```

```
  age ratio gender  frame diabetic
1  46   3.6 female medium        0
2  29   6.9 female  large        1
3  58   6.2 female  large        0
4  67   6.5   male  large        0
5  64   8.9   male medium        0
6  34   3.6   male  large        0
```

(i) (1 pt) Fit a multiple linear regression with ratio as the response and age, gender, and frame as predictors.

```r
mult = lm(ratio ~ age + gender + frame, data = diabetes)
summary(mult)
```

```

Call:
lm(formula = ratio ~ age + gender + frame, data = diabetes)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.9760 -1.1535 -0.2371  0.8712 14.6783 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  4.267848   0.347959  12.265  < 2e-16 ***
```

```
age          0.011385   0.005474   2.080   0.0382 *
gendermale   0.207224   0.175980   1.178   0.2397
framemedium -0.226777   0.213246  -1.063   0.2882
framesmall  -0.970578   0.244779  -3.965 8.75e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.674 on 385 degrees of freedom
Multiple R-squared:  0.07583,	Adjusted R-squared:  0.06623
F-statistic: 7.897 on 4 and 385 DF,  p-value: 3.985e-06
```

(ii) (2 pt) Is there a significant difference in the ratio of HDL between males and females?

There is not a significant difference in the ratio of HDL between males and females because the male coefficient has a p-value of 0.2397 which is greater than $\alpha = 0.1$ and additionally $0 \in [0.207224 - (1.64)0.17598, 0.207224 + (1.64)0.17598]$. This means that the gender male coefficient does not differentiate significantly from baseline (gender = female)

(iii) (2pt) Report and interpret the coefficients for frame.

```r
summary(mult)$coefficients
```

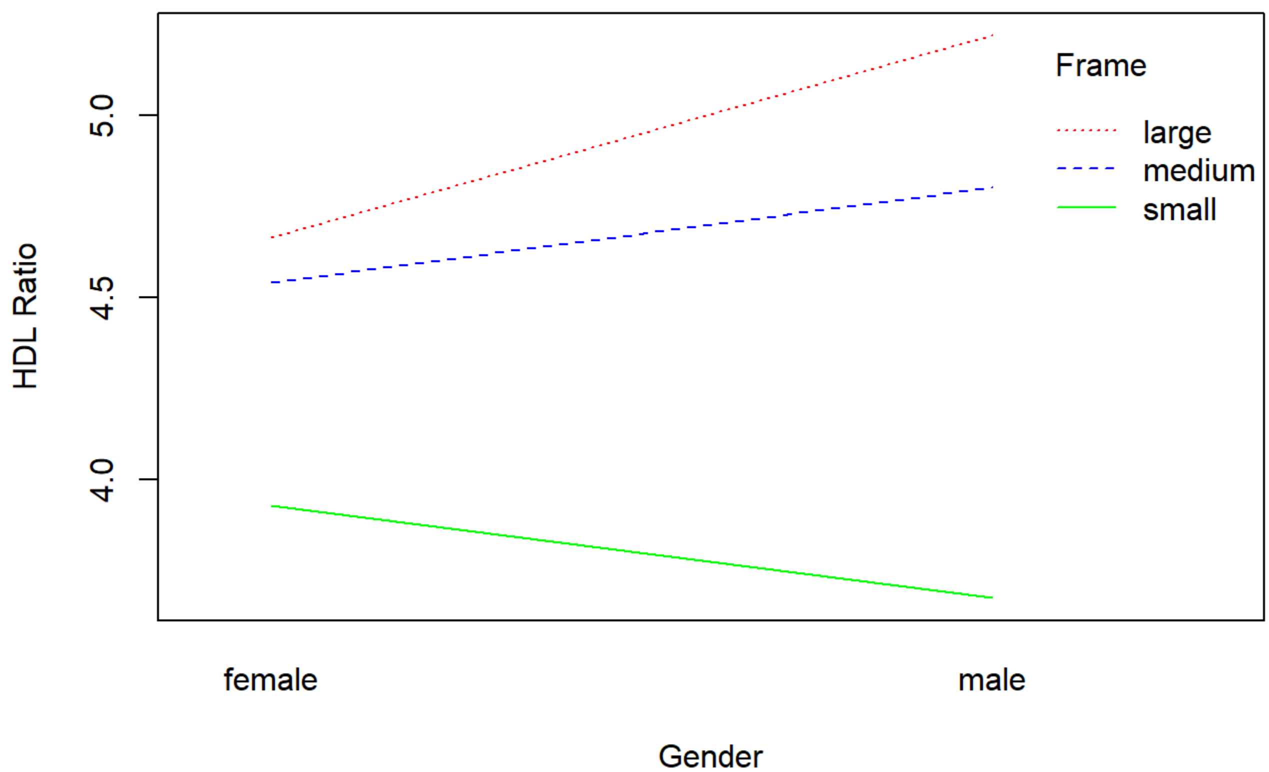
```
               Estimate Std. Error   t value     Pr(>|t|)
(Intercept)  4.26784826 0.34795934 12.265365 2.024249e-29
age          0.01138544 0.00547431  2.079795 3.820548e-02
gendermale   0.20722412 0.17597989  1.177544 2.397057e-01
framemedium -0.22677652 0.21324632 -1.063449 2.882451e-01
framesmall  -0.97057840 0.24477916 -3.965119 8.746412e-05
```

The framesmedium predictor has a coefficients of -0.22677652 with a std of 0.21324632 and p-value of 0.2882451. This means that the difference between a medium frame and a large frame is not statistically significant because the p-value is greater than $\alpha = 0.1$ and $0 \in [-0.22677652 - (1.64)0.21324632, -0.22677652 + (1.64)0.21324632]$.

The framessmall predictor has a coefficients of -0.97057840 with a std of 0.24477916 and p-value of $8.746412e - 05$. This means that the difference between a small frame and a large frame is statistically significant because the p-value is smaller than $\alpha = 0.1$ and $0 \notin [-0.97057840 - (1.64)0.24477916, -0.97057840 + (1.64)0.24477916]$. More specifically, the average change in the response mean from a change from large frame to small frame is -0.97057840.

(iv) (1 pt) Is there an interaction between gender and frame based on the interaction plot?

```
interaction.plot(x.factor = diabetes$gender, trace.factor = diabetes$frame, response=diabetes$r
```



There is an interaction between gender and frame based on the interaction plot because the lines are not parallel. More specifically, it appears there may be an interaction between frame and gender because large/medium frame increases the HDL ratio when going from female to male but the small frame decreases the HDL ration when going from female to male.

(v) (1 pt) If yes, is the interaction effect between gender and frame significant at α “ 0.10?

```
int_model = lm(ratio ~ age + gender * frame + gender + frame, data = diabetes)
summary(int_model)
```

```
Call:
lm(formula = ratio ~ age + gender * frame + gender + frame, data = diabetes)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.0129 -1.1097 -0.2470  0.8891 14.6982 
```

```
Coefficients:
                         Estimate Std. Error t value Pr(>|t|)
(Intercept)              4.081987   0.383829  10.635   <2e-16 ***
age                      0.011213   0.005471   2.050   0.0411 *
gendermale               0.536402   0.335492   1.599   0.1107
framemedium             -0.052085   0.303152  -0.172   0.8637
framesmall              -0.626892   0.331606  -1.890   0.0594 .
gendermale:framemedium -0.277684   0.421682  -0.659   0.5106
gendermale:framesmall  -0.785332   0.485208  -1.619   0.1064
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.673 on 383 degrees of freedom
Multiple R-squared:  0.08229,	Adjusted R-squared:  0.06791
F-statistic: 5.724 on 6 and 383 DF,  p-value: 1.021e-05
```

```
anova(mult, int_model)
```

```
Analysis of Variance Table

Model 1: ratio ~ age + gender + frame
Model 2: ratio ~ age + gender * frame + gender + frame
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    385 1079.1
2    383 1071.6  2    7.5455 1.3484 0.2609
```

The interaction affect appears not to be significant at $\alpha = 0.10$ because the anova table resulted in a p-value of 0.2609 which is greater than our α. This means that the extra predictors do not have statistically significant relationship with the response mean HDL ratio.

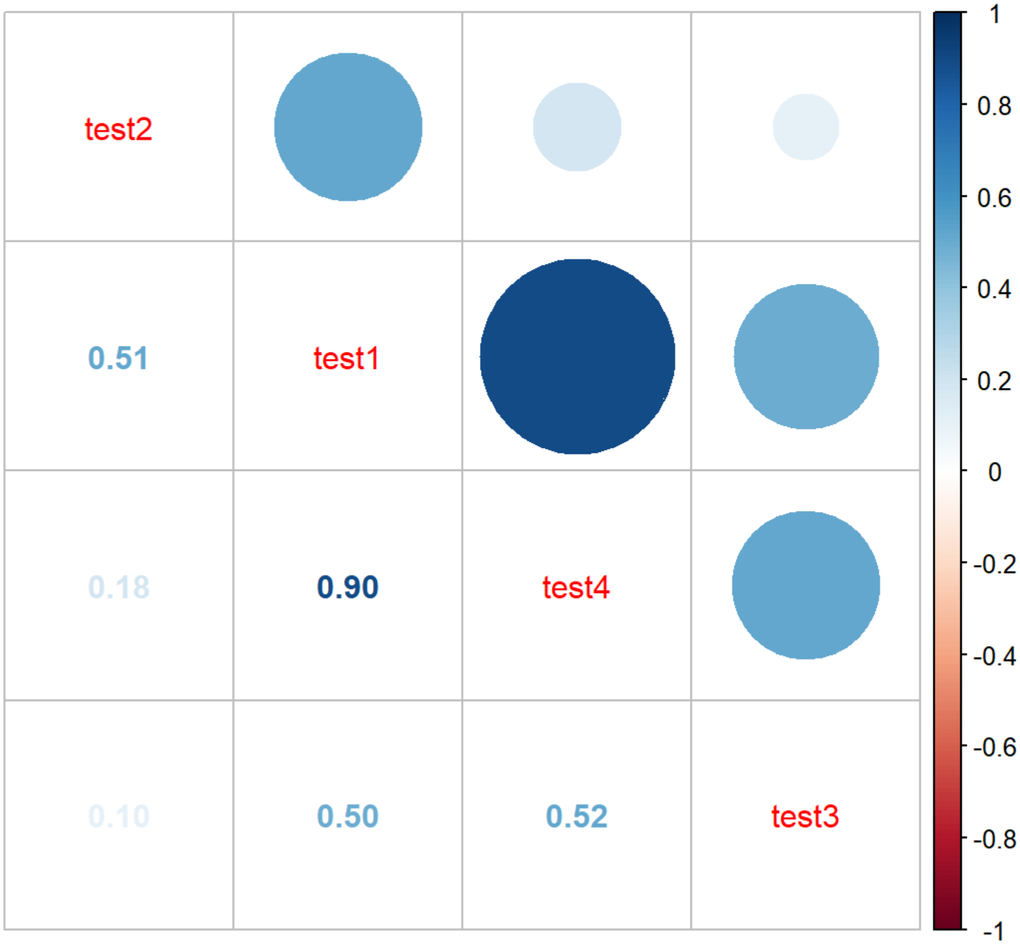
3

```
job_data = read.table("data/jobprof.txt", header=T)

head(job_data)
```

```
  test1 test2 test3 test4 JBS
1    88    86   110   100  87
2    80    62    97    99 100
3    96   110   107   103 103
4    76   101   117    93  95
5    80   100   101    95  88
6    73    78    85    95  84
```

(i) (1 pt) Calculate the correlation matrix of the test scores.

```
X = subset(job_data, select=-c(JBS))
corrplot.mixed(cor(X), order="AOE")
```



(ii) (1 pt) Does the correlation matrix suggest a possible multicollinearity problem for predicting job proficiency score (JPS) from the test scores?

The correlation matrix suggests possible multicollinearity problems because the correlation coefficient between test1 and test4 is 0.9 which is close to 1.

(iii) (1 pt) Fit a multiple linear regression model with JPS as the response and the test scores as the predictors.

```
model = lm(JBS ~ test1 + test2 + test3 + test4, data=job_data)
summary(model)
```

```
Call:
lm(formula = JBS ~ test1 + test2 + test3 + test4, data = job_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-8.7283 -2.6769 -0.7255  3.1096  9.3900 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 100.88142   30.03086   3.359  0.00312 ** 
test1         0.84060    0.21337   3.940  0.00081 ***
test2        -0.19182    0.09142  -2.098  0.04879 *  
test3        -0.04574    0.07258  -0.630  0.53570    
test4        -0.58529    0.40537  -1.444  0.16427    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.212 on 20 degrees of freedom
Multiple R-squared:  0.8014,	Adjusted R-squared:  0.7617 
F-statistic: 20.17 on 4 and 20 DF,  p-value: 8.612e-07
```

(iv) (1 pt) Compute the variance inflation factors for each of the model parameters in (iii).

```
vif(model)
```

```
    test1     test2     test3     test4 
15.172928  3.042143  1.391685 11.384328 
```

(v) (1 pt) What does the VIF suggest?

test1 and test4 have VIF scores above 10 which suggests there is multicollinearity within the data.

(vi) (2 pt) Use a stepwise regression with AIC criterion to select the best model for predicting JPS.

```
library(MASS)   ## this package is required for the stepAIC function
both_sel <- stepAIC(model, direction = "both")
```

```
Start:  AIC=86.97
JBS ~ test1 + test2 + test3 + test4

        Df Sum of Sq    RSS    AIC
- test3  1     10.79 554.10 85.462
```

```
<none>                543.31 86.970
- test4  1     56.63 599.94 87.449
- test2  1    119.59 662.90 89.944
- test1  1    421.65 964.96 99.330

Step:  AIC=85.46
JBS ~ test1 + test2 + test4

        Df Sum of Sq    RSS    AIC
<none>                554.10 85.462
- test4  1     59.74 613.84 86.021
+ test3  1     10.79 543.31 86.970
- test2  1    113.74 667.84 88.129
- test1  1    411.30 965.40 97.342
```